Nonuniform sampling theorems for random signals in the offset linear canonical transform domain

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***Abstract*—With the rapid development of the offset linear canonical transform (OLCT) in the fields of optics and signal processing, it is necessary to consider the nonuniform sampling associated with the OLCT. Nowadays, the analysis and applications of the nonuniform sampling for deterministic signals in the OLCT domain have been well published and studied. However, none of the results about the reconstruction of nonuniform sampling for random signals in the OLCT domain have been proposed until now. In this paper, the nonuniform sampling and reconstruction of random signals in the OLCT domain are investigated. Firstly, a brief introduction to the fundamental knowledge of the OLCT and some special nonuniform sampling models are given. Then, the reconstruction theorems for random signals from nonuniform samples in the OLCT domain have been derived for different nonuniform sampling models. Finally, the simulation results are given to verify the accuracy of theoretical results.**

Keywords: Offset linear canonical transform, Nonuniform sampling, Random signal, Recurrent nonuniform sampling, Power spectral density

I. Introduction

The offset linear canonical transform (OLCT) [1,2,3,5], also known as the special affine Fourier transformation [1] or the inhomogeneous canonical transform [2], is a six-parameter class of linear integral transform. This generalized transform can be interpreted as a time shifting and frequency modulated version of the linear canonical transform (LCT), with more flexible and general for its two extra parameters u_0 and w_0. Many well-known signal processing operations, such as the Fourier transform (FT), the fractional Fourier transform (FRFT) [4,6], the Fresnel transform (FRT) [8], the LCT [7,11,17], time shifting and scaling, frequency modulation and many others are all its special cases. In recent years, the OLCT has been found various applications in optics and signal processing community [14,21]. Therefore, it is theoretically worthwhile and practically useful to develop relevant theorems for the OLCT.

The sampling process is one of the most fundamental concepts of digital signal processing, which serves as a bridge between the continuous physical signals and the discrete domain. The sampling theorem expansions for signals band-limited in the FRFT domain, in the LCT domain and in the OLCT domain have been presented in [3,9-19], which assume that the signal at hand is obtained by uniform sampling an analog signal with one sampling channel. However, all those sampling theorems are dealt with the deterministic signals. In some real applications, one often encounters signals that do not have a precise mathematical description, which are called random signals, since they develop as random functions of time. It is natural to ask what the sampling theorems associated with random signals obey.

As far as we know, the nonuniform sampling of random signals has various practical applications. For example, nonuniform sampling can be encountered in various data acquisition systems due to imperfect sampling timebase, and the errors it generates are usually dominating and cannot be neglected in precision instrumentation[16,17]. However, there are no research papers published associated with the reconstruction of the random signal from various nonuniform samples in the OLCT domain. Therefore, it is worthwhile and practically useful to derive nonuniform sampling theorems for random signals in the OLCT domain.

The purpose of this paper is to propose nonuniform sampling theorems for random signals in the OLCT domain. The outline of this paper is organized as follows. The preliminaries are proposed in next section. In Section 3, the nonuniform sampling theorems for random signals in the OLCT domain are obtained. The simulation results and potential applications are presented to show that the theories are accurate and useful in Section 4. Finally, Section 5 summarizes and concludes this paper.

II. Preliminaries

A. The offset linear canonical transform

The OLCT of a signal $f(t)$ with real parameters $A = (a, b, c, d, u_0, w_0)$ is defined as

$$OL_f^A[f(t)](u) = \begin{cases} \int_R f(t)K_A(u,t)\mathrm{d}t, & b \neq 0 \\ \sqrt{d}e^{j\frac{cd(u-u_0)^2}{2}+jw_0u} f[d(u-u_0)], & b = 0 \end{cases} \tag{1}$$

where

$$K_A(u,t) = \frac{1}{\sqrt{j2\pi b}} e^{\frac{j}{2b}[at^2+2t(u_0-u)-2u(du_0-bw_0)+du^2+du_0^2]} \tag{2}$$

and a, b, c, d, u_0, w_0 are real numbers satisfying $ad - bc = 1$.

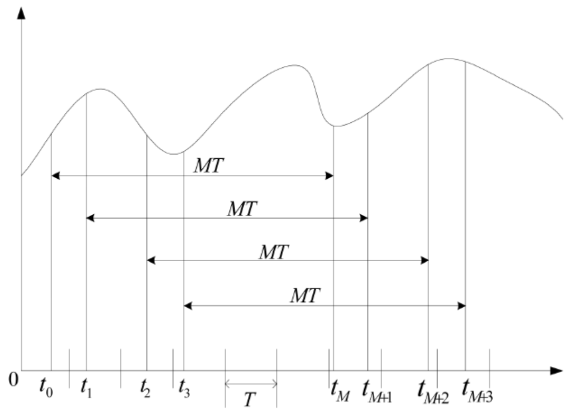

Fig. 1. The periodic nonuniform sampling model

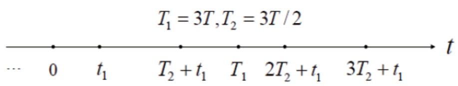

Fig. 2. Nth-order recurrent nonuniform sampling model

Note that when $b = 0$, the OLCT becomes a chirp multiplication. Therefore, we only consider the case of $b \neq 0$. For further details about properties of the OLCT, readers can refer to [1,2].

B. Nonuniform sampling models

1) The periodic nonuniform sampling model: The periodic nonuniform sampling is a special case of nonuniform sampling, which is also known as block sampling or recurrent nonuniform sampling [3]. The model of this periodic nonuniform sampling distribution is shown in Fig.1. In this recurrent nonuniform sampling model, the sampling points are divided into several groups of points. The groups have a recurrent period NT, and each group has N nonuniform sampling points. Denoting the points in one period by t_p, $p = 1, 2, ..., N$, the complete set of sampling can be marked as $\tau_p m = tp + mNT, m = ..., 1, 0, 1, ...$

2) Nth-order recurrent nonuniform sampling model: Nth-order recurrent nonuniform sampling is another recurrent nonuniform sampling [26]. An example of this recurrent nonuniform sampling model is depicted in Fig.2. In this form of sampling, the recurrent nonuniform samples can be regarded as a combination of N sequences of uniform samples taken at one Nth of the Nyquist rate, the set of sampling points are $t_p + nT_p, p = 1, ..., N, n \in (-\infty, \infty)$, where $\sum_{p=1}^{N} \frac{1}{T_p} = \frac{1}{T}$.

3) Nonuniform sampling due to migration of a finite number of uniform samples model: One of the simplest nonuniform

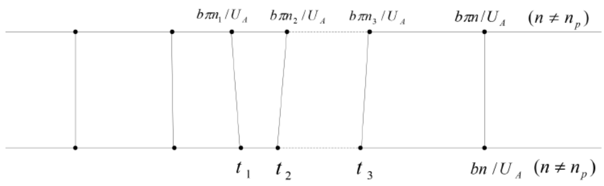

Fig. 3. Nonuniform sampling due to migration of a finite number of uniform samples model

sample point distributions is obtained from the migration of a finite number of sample points in a uniform distribution. This kind of nonuniform sampling model can be modeled as shown in Fig.3 [3]. In this kind of uniform sampling model, N uniform sample points located at $t = \frac{bn_p\pi}{U_A}$, where n_p with $p = 1, 2, ..., N$ are N distinct integers, have been migrated to N new positions, and $t = t_p$, and $\frac{U_A t_p}{\pi b}$ is not an integer.

C. The OLCT power spectral density

In traditional FT domain, for a random signal $x(t)$, we often use the autocorrelation function and the power spectral density to express its characters. Motivated by the method in [4] for the fractional Fourier transform and the definition in LCT by [11], we generalize the definitions of the OLCT auto-correlation function and the OLCT power spectral density in this subsection that are used to efficiently characterize a random signal in the OLCT domain.

Theorem 2.1: Let $x(t)$ be a random signal. Then, denote the OLCT auto-correlation function of $x(t)$ as

$$R_{xx}^{A}(\tau) = \lim_{T \to +\infty} \frac{1}{2T} \int_{-T}^{T} R_{xx}(t + \tau, t) e^{\frac{j\tau}{b}(at+u_0)} dt \tag{3}$$

Likewise, the OLCT auto-power spectral density of $x(t)$ is given below:

$$P_{xx}^{A}(u) = \sqrt{\frac{1}{-j2\pi b}} OL_A\{R_{xx}^{A}(\tau)\}(u) e^{\frac{-j}{2b}(du^2 - 2du_0u + 2bw_0u)} \tag{4}$$

where $R_{xx}(t + \tau, t)$ is the traditional auto-correlation function of $x(t)$.

Then a signal $x(t)$ is called U_A band-limited signal in the OLCT domain with parameter A, which means its power spectral density satisfies

$$P_{xx}^{A}(u) = 0 \quad for\, |u| > U_A \tag{5}$$

where U_A is called the bandwidth of signal $x(t)$ in the OLCT domain.

Similar conventions are widely used in the study of the sampling of random signals, where $x(t)$ is assumed to be stationary in a wide sense for which the auto-correlation function is square integrable and band limited. In this paper, we will focus on $x(t)$ such that $x(t)e^{\frac{j}{2b}(at^2+2u_0t)}$ is stationary in a wide sense. In this case, $R_{xx}(t + \tau, t)e^{\frac{j\tau}{b}(at+u_0)}$ does not depend on t. Therefore, $R_{xx}^{A}(\tau)$ is well defined.

III. Nonuniform sampling theorems for random signals in the OLCT domain

Some special nonuniform sampling models have been presented in previous section. These nonuniform sampling models occur in many practical applications. Theories and applications of these special nonuniform sampling models for deterministic signal in the OLCT domain have been well studied in [5,8]. However, there are no papers focus on the nonuniform sampling theorems for random signals in the OLCT domain. Therefore, the problem of reconstructing the random signals from nonuniform samples will be considered in this section. First, the periodic nonuniform sampling theorem for random signals in the OLCT domain has been derived. Then, we formulate the Nth-order recurrent nonuniform sampling theorem for random signals in the OLCT domain. Subsequently, the reconstruction of the random signal from the migration of a finite number of sample points in a uniform distribution in the OLCT domain has been presented. Finally, the general nonuniform sampling theorem for random signals associated with the OLCT also has been obtained.

A. The periodic nonuniform sampling theorem

The periodic nonuniform sampling theorem for a deterministic signal bandlimited in the OLCT domain has been investigated in [7]. In what follows, the reconstruction formula for a random signal bandlimited in the OLCT domain from its recurrent nonuniform samples is proposed.

Theorem 3.1: Assume a random signal $x(t)$ which is bandlimited to U_A in the OLCT sense with parameter A. If its chirped form, $x(t)e^{\frac{j}{2b}(at^2+2u_0t)}$, is stationary in the wide sense, then $x(t)$ can be reconstructed from its nonuniformly sampled points $\tau_{pm}=t_p+mNT, m=...-1,0,1,...$ in the mean square sense:

$$x(t)=l.i.m.e^{-\frac{j}{2b}(at^2+2u_0t)} \times\sum_{m=-\infty}^{\infty}\sum_{p=1}^{N}x(\tau_{pm})e^{\frac{j}{2b}(a\tau_{pm}^2+2u_0\tau_{pm})}\psi_{pm}(t) \tag{6}$$

where $\psi_{pm}(t)=\frac{\prod_{q=1}^{N}sin(\frac{U_A}{bN}(t-t_q))}{\prod_{q=1,q\neq p}^{N}sin(\frac{U_A}{bN}(t_p-t_q))}\frac{(-1)^{mN}}{\frac{U_A}{bN}(t_p-t_q-mN\frac{\pi b}{U_A})}$ and $l.i.m.$ denotes limit in the mean square sense or convergence in probability as well, i.e.,

$$\lim_{N\to+\infty}E[|x(t)-e^{-\frac{j}{2b}(at^2+2u_0t)} \times\sum_{m=-\infty}^{\infty}\sum_{p=1}^{N}x(\tau_{pm})e^{\frac{j}{2b}(a\tau_{pm}^2+2u_0\tau_{pm})}\psi_{pm}(t)|]^2=0 \tag{7}$$

Proof. Let the estimate $\widetilde{x}(t)$ be

$$x(t)=e^{-\frac{j}{2b}(at^2+2u_0t)} \times\sum_{m=-\infty}^{\infty}\sum_{p=1}^{N}x(\tau_{pm})e^{\frac{j}{2b}(a\tau_{pm}^2+2u_0\tau_{pm})}\psi_{pm}(t) \tag{8}$$

Then, we have

$$E\{[x(t)-\widetilde{x}(t)]x^*(t)\}=R_{xx}(t,t)-e^{\frac{-j}{2b}(at^2+2u_0t)} \times\sum_{m=-\infty}^{\infty}\sum_{p=1}^{N}R_{xx}(\tau_{pm},t)e^{\frac{j}{2b}(a\tau_{pm}^2+2u_0\tau_{pm})}\psi_{pm}(t) \tag{9}$$

Since $e^{\frac{j}{2b}(at^2+2u_0t)}$ is stationary in the wide sense, we derive that

$$E\left[x(t_1)e^{\frac{j}{2b}(at_1^2+2u_0t_1)}x^*(t_2)e^{\frac{-j}{2b}(at_2^2+2u_0t_2)}\right] =e^{\frac{j}{2b}(a\tau^2+2a\tau t_2+2u_0\tau)}R_{xx}(t_2+\tau,t_2) \tag{10}$$

is only a function of the variable τ where $\tau=t_1-t_2$. Therefore, $e^{\frac{j}{2b}(a\tau^2+2a\tau t_2+2u_0\tau)}R_{xx}(t_2+\tau,t_2)$ is only a function of the variable τ. So the OLCT autocorrelation function of $x(t)$ can be expressed as

$$\begin{aligned}R_{xx}^A(\tau)&=\lim_{T\to+\infty}\frac{1}{2T}\int_{-T}^{T}R_{xx}(t_2+\tau,t_2)e^{\frac{j\tau}{b}(at_2+u_0)}dt_2\\&=R_{xx}(\rho+\tau,\rho)e^{\frac{j}{b}(a\rho+u_0)\tau}\end{aligned} \tag{11}$$

where the equation is valid for all ρ. Substituting (11) into (9), we deduce that

$$E\{[x(t)-\widetilde{x}(t)]x^*(t)\}=R_{xx}^A(0)-e^{-j\frac{(at^2+2u_0t)}{2b}} \times\sum_{m=-\infty}^{\infty}\sum_{p=1}^{N}R_{xx}^A(\tau_{pm}-t)e^{\frac{j}{2b}(a\tau_{pm}^2-2at\tau_{pm}+2at^2+2u_0t)}\psi_{pm}(t) \tag{12}$$

Meanwhile, according to the space shift and phase shift properties in the OLCT domain, we can easily obtain that

$$\begin{aligned}&L_A[R_{xx}^A(\lambda-\lambda_0)e^{\frac{-j\lambda}{b}(a\lambda_0+u_0)}](u)\\&=e^{\frac{-ja\lambda_0^2-ju_0\lambda_0}{2b}-\frac{j\lambda_0u}{b}}L_A[R_{xx}^A(\lambda)](u)\\&=\sqrt{-j2\pi b}P_{xx}^A(u)e^{\frac{j}{2b}(du^2-2du_0u+2bw_0u-a\lambda_0^2-u_0\lambda_0-2\lambda_0u)}\end{aligned} \tag{13}$$

Noting that the random signal $x(t)$ is bandlimited in the OLCT domain, so the shifted correlation function $R_{xx}^A(\lambda-\lambda_0)e^{\frac{-j\lambda}{b}(a\lambda_0+u_0)}$ is bandlimited in the OLCT domain with bandwidth U_A. Then, by applying the nonuniform sampling theorem to the deterministic function $R_{xx}^A(\lambda-\lambda_0)e^{\frac{-j\lambda}{b}(a\lambda_0+u_0)}$, we derive

$$R_{xx}^A(\lambda-\lambda_0)e^{\frac{-j\lambda}{b}(a\lambda_0+u_0)}=e^{\frac{-j}{2b}(a\lambda^2+2u_0\lambda)} \times\sum_{m=-\infty}^{\infty}\sum_{p=1}^{N}R_{xx}^A(\tau_{pm}-\lambda_0)e^{\frac{ja}{2b}(\tau_{pm}^2-2\lambda\tau_{pm})}\psi_{pm}(\lambda) \tag{14}$$

Let the variables $\lambda=\lambda_0=t$ in (14), we have

$$R_{xx}^A(0)e^{\frac{-j}{b}(at^2+u_0t)}=e^{\frac{-j}{2b}(at^2+2u_0t)} \sum_{m=-\infty}^{\infty}\sum_{p=1}^{N}R_{xx}^A(\tau_{pm}-t)e^{\frac{ja}{2b}(\tau_{pm}^2-2t\tau_{pm})}\psi_{pm}(t) \tag{15}$$

Then substituting (15) into (12), we finally obtain

$$E\{[x(t)-\widetilde{x}(t)]x^*(t)\}=0 \tag{16}$$

On the other hand,

$$\begin{aligned} &E\{[x(t)-\widetilde{x}(t)]x^*(\tau_{pn})\} = R_{xx}(t,\tau_{pn}) - e^{\frac{-j}{2b}(at^2+2u_0t)} \\ &\times \sum_{m=-\infty}^{\infty}\sum_{p=1}^{N} R_{xx}(\tau_{pm},\tau_{pn}) e^{\frac{j}{2b}(a\tau_{pm}^2+2u_0\tau_{pm})}\psi_{pm}(t) \\ &= R_{xx}^A(t-\tau_{pm}) e^{\frac{-j}{b}(a\tau_{pn}+u_0)(t-\tau_{pm})} - e^{-j\frac{(at^2+2u_0t)}{2b}} \\ &\times \sum_{m=-\infty}^{\infty}\sum_{p=1}^{N} R_{xx}^A(\tau_{pm}-\tau_{pn}) \\ &\times e^{\frac{j}{2b}(a\tau_{pm}^2-2a\tau_{pn}\tau_{pm}+2a\tau_{pn}^2+2u_0\tau_{pn})}\psi_{pm}(t) \end{aligned} \tag{17}$$

Interchanging the variables $\lambda = t, \lambda_0 = \tau_{pn}$ in (14), we have

$$\begin{aligned} &R_{xx}^A(t-\tau_{pn}) e^{\frac{-jt}{b}(a\tau_{pn}+u_0)} = e^{-j\frac{(at^2+2u_0t)}{2b}} \\ &\sum_{m=-\infty}^{\infty}\sum_{p=1}^{N} R_{xx}^A(\tau_{pm}-\tau pn) e^{\frac{ja}{2b}(\tau_{pm}^2-2\tau_{pn}\tau_{pm})}\psi_{pm}(t) \end{aligned} \tag{18}$$

Then, combining (17) with (18), we deduce

$$E\{[x(t)-\widetilde{x}(t)]x^*(\tau_{pn})\} = 0 \tag{19}$$

Since $\widetilde{x}(t)$ is a linear summation of the $x(\tau_{pm})$ in (7), we can get

$$E[x(t)-\widetilde{x}(t)]\widetilde{x}^*(t) = 0 \tag{20}$$

Therefore, it follows from (16) and (20) that

$$\begin{aligned} E\left[|x(t)-\widetilde{x}(t)|^2\right] &= E\left[(x(t)-\widetilde{x}(t))(x^*(t)-\widetilde{x}^*(t))\right] \\ &= E\left[(x(t)-\widetilde{x}(t))x^*(t)\right] E\left[(x(t)-\widetilde{x}(t))\widetilde{x}^*(t)\right] \\ &= 0 \end{aligned} \tag{21}$$

This concludes the proof of Theorem 1.

B. Nth-order recurrent nonuniform sampling

In this subsection, we consider a subclass of recurrent nonuniform sampling, i.e., Nth order recurrent nonuniform sampling, in which the sampling points in each period can be further divided into groups with a common inner period [28]. For this kind of nonuniform sampling, the following theorem for random signals in the OLCT domain is valid.

Theorem 3.2: Let a random signal $x(t)$ be bandlimited to U_A in the OLCT domain with parameter A. If its chirped form, $x(t)e^{\frac{j}{2b}(at^2+2u_0t)}$, is stationary in the wide sense, then $x(t)$ can be reconstructed by the Nth-order recurrent nonuniformly sampled points $\tau_p = t_p + nT_p$ in the mean square sense:

$$\begin{aligned} &x(t) = l.i.m. e^{-\frac{j}{2b}(at^2+2u_0t)} \\ &\times \sum_{n=-\infty}^{\infty}\sum_{p=1}^{N} x(\tau_p) e^{\frac{j}{2b}(a\tau_{pn}^2+2u_0\tau_{pn})}\psi_{pn}(t) \end{aligned} \tag{22}$$

where $\psi_{pn}(t) = \dfrac{T_p\prod_{q=1}^{N}\sin(\frac{\pi(t-t_p)}{T_q})}{\pi(-1)^n\prod_{q=1,q\neq p}^{N}\sin(\frac{\pi(t-t_p)}{T_q}+n\pi\frac{T_p}{T_q})(t-nT_p-t_p)}$ and $l.i.m.$ denotes limit in the mean square sense or convergence in probability as well, i.e.,

$$\begin{aligned} &\lim_{N\to+\infty} E[|x(t) - e^{-\frac{j}{2b}(at^2+2u_0t)} \\ &\times \sum_{n=-\infty}^{\infty}\sum_{p=1}^{N} x(\tau_{pn}) e^{\frac{j}{2b}(a\tau_{pn}^2+2u_0\tau_{pn})}\psi_{pn}(t)|]^2 = 0 \end{aligned} \tag{23}$$

Proof. Similar to the proof of Theorem 1.

C. Nonuniform sampling due to migration of a finite number of uniform samples

The migration of a finite number of sample points in a uniform distribution is one of the simplest nonuniform sampling distribution [16]. In this subsection, we study the sampling theorem for random signals associated with the OLCT in this kind of nonuniform sampling model. Then we have the following theorem.

Theorem 3.3: Let a random signal $x(t)$ be bandlimited to U_A in the OLCT domain with parameter A. If its chirped form, $x(t)e^{\frac{j}{2b}(at^2+2u_0t)}$, is stationary in the wide sense, then $x(t)$ can be reconstructed by the Nth-order recurrent nonuniformly sampled points τ_m in the mean square sense:

$$\begin{aligned} &x(t) = l.i.m. e^{-\frac{j}{2b}(at^2+2u_0t)} \\ &\sum_{m=-\infty}^{\infty} x(\tau_p) e^{\frac{j}{2b}(a\tau_m^2+2u_0\tau_m)}\psi_m(t) \end{aligned} \tag{24}$$

where

$$\psi_m(t) = \begin{cases} \dfrac{\prod_{q=1}^{N}(t-t_q)\prod_{q=1}^{N}\frac{b\pi}{U_A}(n-n_q)}{\prod_{q=1}^{N}(t-\frac{n_q\pi b}{U_A})\prod_{q=1}^{N}(\frac{n\pi b}{U_A}-t_q)}\dfrac{(-1)^n b\sin(\frac{U_At}{b})}{U_At-n\pi b} & \tau_m = \frac{n\pi b}{U_A} \neq \frac{n_q\pi b}{U_A} \\ \dfrac{\prod_{q=1,q\neq p}^{N}(t-t_q)\prod_{q=1}^{N}(t_q-\frac{n_q\pi b}{U_A})}{\prod_{q=1,q\neq p}^{N}(t_p-t_q)\prod_{q=1}^{N}(t-\frac{n_q\pi b}{2U_A})}\dfrac{\sin(\frac{U_At}{b})}{\sin(\frac{U_At_p}{b})} & \tau_m = t_p \end{cases} \tag{25}$$

and $l.i.m.$ denotes limit in the mean square sense or convergence in probability as well, i.e.,

$$\begin{aligned} &\lim_{N\to+\infty} E[|x(t) - e^{-\frac{j}{2b}(at^2+2u_0t)} \\ &\sum_{m=-\infty}^{\infty} x(\tau_m) e^{\frac{j}{2b}(a\tau_m^2+2u_0\tau_m)}\psi_m(t)|]^2 = 0 \end{aligned} \tag{26}$$

Proof. Similar to the proof of Theorem 1.

D. The general nonuniform sampling

In this subsection, the general nonuniform sampling model is presented. This type of nonuniform sampling pattern is considered and a classical reconstruction theorem for bandlimited deterministic signals in the LCT domain is deduced[28]. Similar to the deterministic signal case, we study the corresponding result for random signals associated with the OLCT in the following:

Theorem 3.4: Suppose a random signal $x(t)$ is bandlimited to U_A in the OLCT domain with parameter A. If its chirped form, $x(t)e^{\frac{j}{2b}(at^2+2u_0t}$, is stationary in the wide sense,

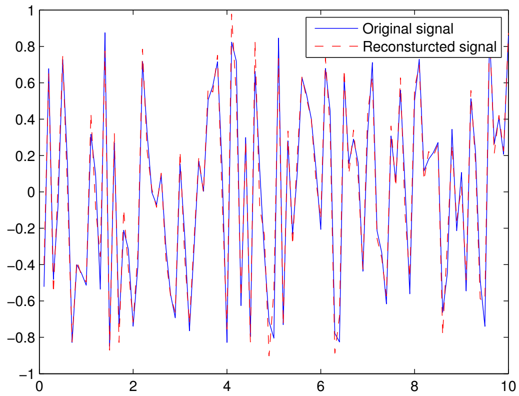

Fig. 4. The reconstructed signal with the periodic nonuniform sampling

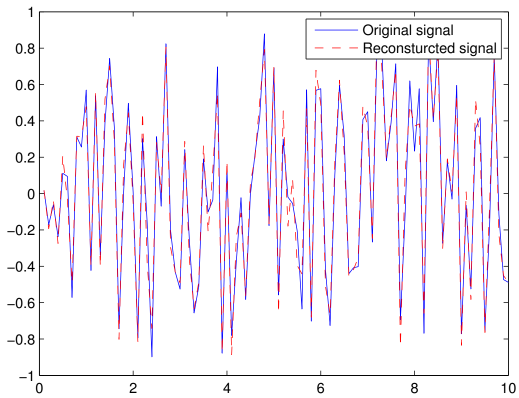

Fig. 5. The reconstructed signal with the periodic nonuniform sampling

then $x(t)$ is uniquely determined by its samples $x(t_n)$ if $\left|t_n - n\frac{b\pi}{U_A}\right| \leq D \leq \frac{b\pi}{4U_A}$ and the reconstruction theorem can be expressed as:

$$x(t) = l.i.m.e^{-\frac{j}{2b}(at^2+2u_0t)} \times \sum_{n=-\infty}^{\infty} x(\tau_p)e^{\frac{j}{2b}(at_n^2+2u_0t_n)}\frac{G(t)}{G'(t_n)(t-t_n)} \tag{27}$$

where $G(t) = e^{\alpha t}(t-t_0)\prod_{n\neq 0}\frac{1-t}{t_n}e^{\frac{t}{t_n}}$; $\alpha = \sum_{n\neq 0}\frac{1}{t_n}$, $D \in R$; and $G^{'}(t_n)$ is the derivative of $G(t)$ evaluated at $t = t_n$ and $t_n = 0$ for some n, when $t_0 = 0$.

Proof. Similar to the proof of Theorem 1.

IV. Simulations

Although the proposed formula is derived theoretically, we will still give a simple numerical example to illustrate the validity of the method and show the advantage of using the OLCT. Simulation tests are implemented with MATLAB 2016a.

In the simulation, The nominal sampling period T = 0.1 and the parameters $A = [1, 1.8, 0, 1, 2, 0]$. By applying the reconstruction theorem of the recurrent nonuniform sampling, the reconstruction signal of $x(t)$ can be obtained in Fig.4. And Fig.5. is the the reconstruction signal using the Nth-order recurrent nonuniform sampling theorem. It is easy to see that the simulation results are coinciding with the theorems we derived very well.

V. Conclusions

In this paper, we have investigated the nonuniform sampling and reconstruction of random signals associated with the OLCT in the mean square sense. First, some special nonuniform sampling models have been introduced and the OLCT power spectral density is given. Then, the reconstruction formulas for random signals in the OLCT domain have been derived based on these models. Finally, the simulation results and the potential applications are proposed to verify the results. It shows that the methods presented in this paper are correct and effective. Further research and work will be in the direction of these theories for nonstationary random signals and applications of these results in the interleaved A/D converters for the nonstationary signals in the OLCT domain.

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References

[1] S. Abe, J. T. Sheridan, "Optical operations on wave functions as the Abelian Subgroups of the special affine Fourier transformation," *Opt. Lett.* vol.19(22) pp.1801-1803. November 1994

[2] S. C. Pei, J. J. Ding, "Eigenfunctions of the offset Fourier,fractional Fourier, and linear canonical transforms," *J.Opt.Soc.Am.A Opt.Image Sci.Vis.* vol.20(3). pp.522-532. March 2003

[3] S. Q. Xu, Y. Chai, Y. Q. Hu, C. M. Jiang, Y. Li, "Reconstruction of digital spectrum from periodic nonuniformly sampled signals in offset linear canonical transform domain," *Opt. Commun.* vol. 348, pp. 59-65. Augest 2015

[4] R. Tao, F. Zhang, Y. Wang, "Fractional Power Spectrum," *IEEE. Trans. Signal Process.* vol. 56(9) pp.4199-4206. April 2008

[5] Q. Xiang, K. Y. Qin, "Convolution,correlation,and sampling theorems for the offset linear canonical transform," *Signal Image Video Process.* vol. 8(3), pp. 433-442. March 2014

[6] L. Almeida, "The fractional Fourier transform and time-frequency representations," *IEEE Trans.Signal Process.* vol. 42(11), pp. 30843091, 1994

[7] D. Y. Wei, Q. W. Ran, Y. M. Li, "Reconstruction of band-limited signals from multichannel and periodic nonuniform samples in the linear canonical transform domain," *Opt. Commun.* vol. 284(19), pp. 4307-4315. September 2011

[8] D. F. V. James, G. S. Agarwal, "The generalized Fresnel transform and its applications to optics," *Opt. Commun.*, vol.126 (4), pp. 207-212, May 1996.

[9] R. Tao, F. Zhang, Y. Wang, "Sampling random signals in a fractional Fourier domain,"*Signal Process.* vol. 91(6), pp.1394-1400. June 2011.

[10] Q. Xiang, K. Y. Qin, Q. Z, Huang, "Multichannel Sampling of Signals Band-Limited in Offset Linear Canonical Transform Domains,"*Circuits Syst. Signal Process.* vol. 32(5), pp. 2385-2406. October 2013.

[11] H. Y. Huo, W. C. Sun, "Sampling theorems and error estimates for random signals in the linear canonical transform domain,"*Signal Process.* vol.111, pp. 31-38. June 2015.

[12] R. Tao, B. Z. Li, Y. Wang, "On sampling of band-limited signals associated with the linear canonical transform," *IEEE Trans. Signal Process.* vol. 56(11), pp. 5454-5464. November 2008.

[13] B.Z. Li, R. Tao, Y. Wang, “New sampling formulae related to linear canonical transform,”*Signal Process.* vol.87(5), pp.983-990. May 2007.

[14] D. Y. Wei, Y. M. Li, “Sampling reconstruction of N-dimensional bandlimited images after multilinear filtering in fractional Fourier domain,” *Opt. Commun.* vol. 295, pp. 26-35, May 2013.

[15] X. Li, W. C. Sun, “Sampling theorems for signals periodic in the linear canonical transform domain,” *Opt. Commun.* vol.290, pp. 14-18, March 2013.

[16] Y. C. Jenq, “Digital spectra of nonuniformly sampled signals: fundamentals and high-speed waveform digitizers,”*IEEE Trans. Instrum. Meas.* vol.37(2) pp. 245-251, June 1988.

[17] B. Z. Li, T. Z. Xu, “Sampling in the linear canonical transform domain,”*Math.Probl. Eng.* pp.1-12, 2012.

[18] J. Yen,“On nonuniform sampling of bandwidth-limited signals,” *IRE Trans. Circuit Theory* vol 3(4), pp.251-257, 1957.

[19] A. Stern, “Sampling of compact signals in the offset linear canonical transform domains,” *Signal Image Video Process.* vol 1(4), pp.359-367, 2007.

[20] S. C. Pei, J. J. Ding, “Eigenfunctions of Fourier and fractional Fourier transforms with complex offsets and parameters,”*IEEE Trans.Circuits Syst.I* vol. 54(7) pp. 1599-1611, 2007

[21] H. Zhao, R. Y. Wang, D. P. Song, “Recovery of bandlimited signals in linear canonical transform domain from noisy samples” *Circuits Syst. Signal Process.* vol.33(6), pp. 1997-2008, June 2014

[22] S. Q. Xu, Y. Chai, Y. Q. Hu, “Spectral analysis of sampled band-limited signals in the offset linear canonical transform domain,”*Circuits Syst. Signal Process.* vol.34(12), pp. 3979-3997, December 2014.

[23] R. Tao, B. Z. Li, Y. Wang, “Spectral analysis and reconstruction for periodic nonuniformly sampled signals in fractional Fourier domain,” *IEEE Trans. Signal Process.*, vol.55(7), pp. 3541-3547, July 2007.

[24] Y. J. Cao, B. Z. Li, Y. H. Chen, “Logarithmic uncertainty relations for odd or even signals associate with Wigner-Ville distribution,” *Circuits Syst. Signal Process.* vol.35(7), pp.1338-1353, July 2016.

[25] B. Z. Li, R. Tao, Y. Wang, “New sampling formulae related to linear canonical transform” *Signal Process.* vol. 87(5), pp. 983-990, May 2007.

[26] F. Zhang, Y. Hu, R. Tao, “Signal reconstruction from partial information of discrete linear canonical transform,”*Opt. Eng.* vol.53 (3), March 2014.

[27] J. P. Yan, B. Z. Li, Y. H. Chen, Q. Y. Cheng, “Distribution moments associated with the linear canonical transform,” *Int. J. Electron.*,vol.100 (4), pp. 473-481, 2013

[28] Y. C. Eldar, A. V. Oppenheim, “Filterbank reconstruction of bandlimited signals from nonuniform and generalized samples,”*IEEE Trans. Signal Process.*, vol. 48(10), pp.2864-2875, October 2000.